Cloud, Microservices and Applications
Lab Assignment 3

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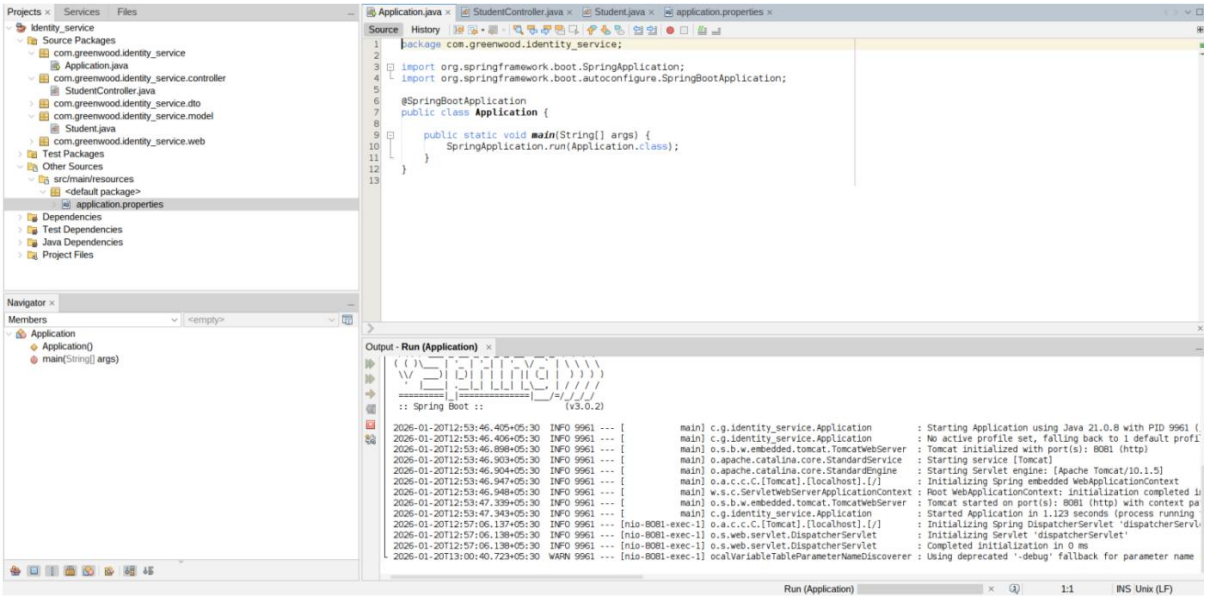
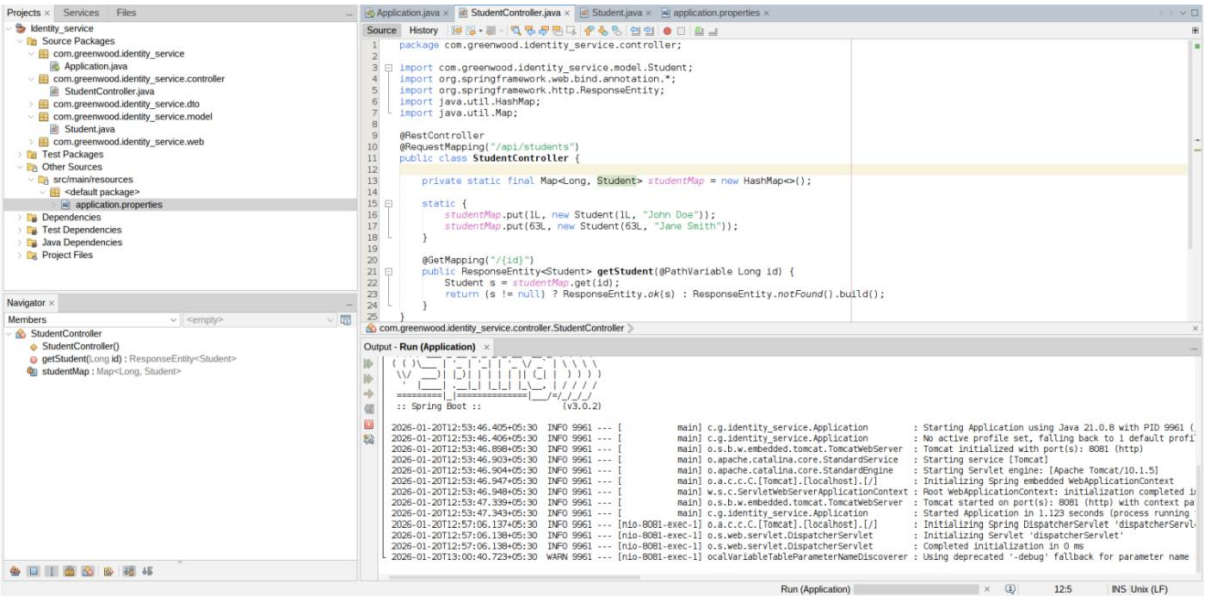
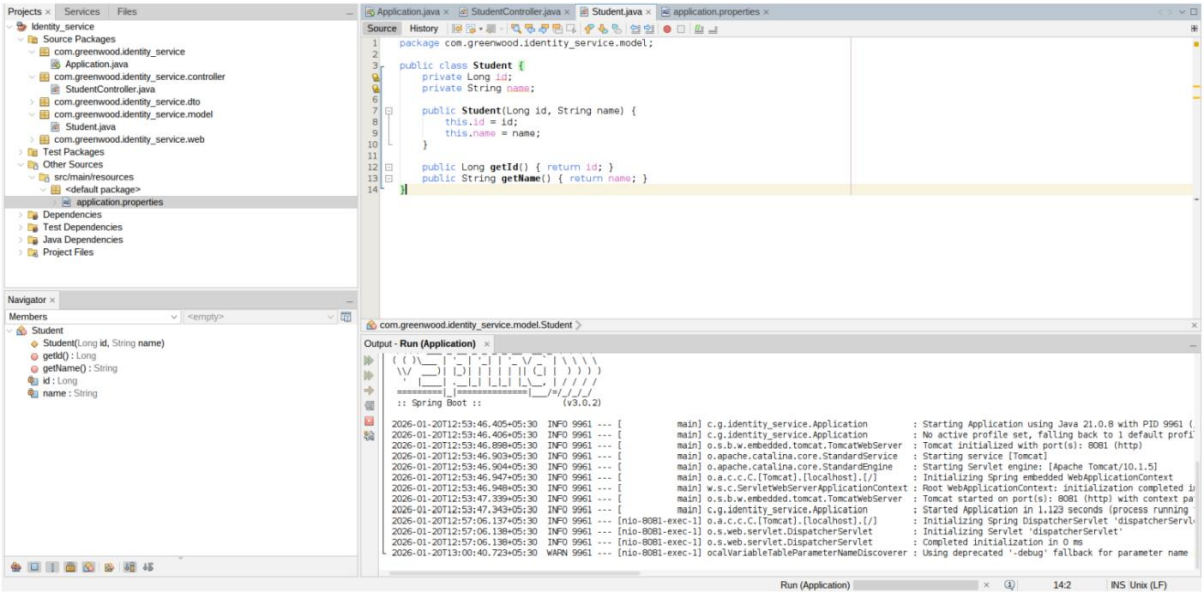
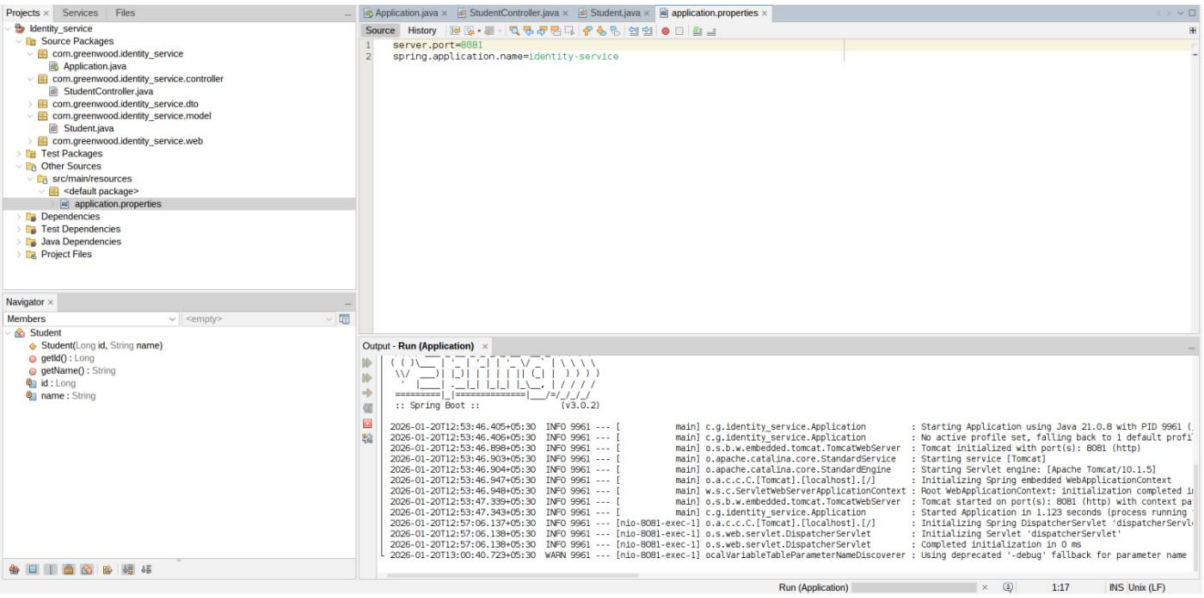
Setting up 2 interconnected microservices and running them on Docker

Microservice 1 (Identity service):

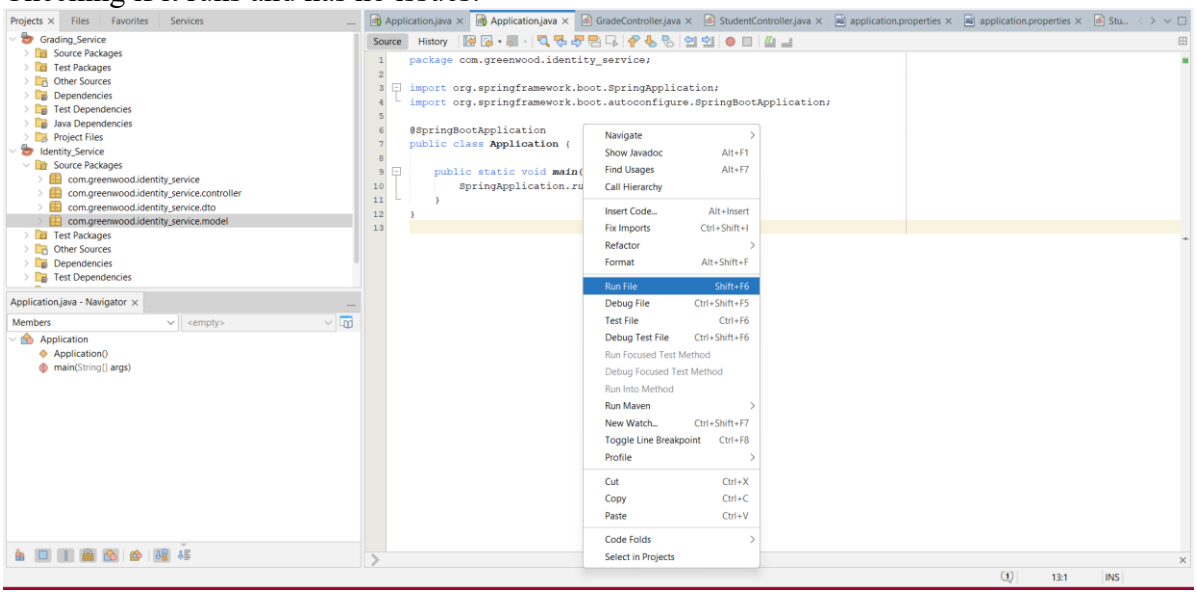


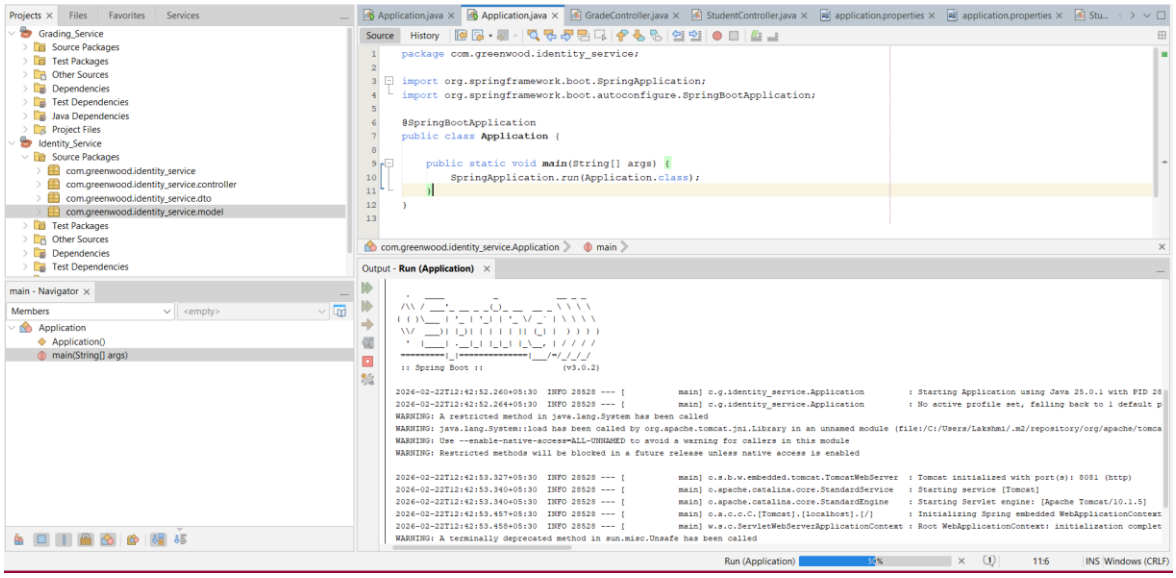




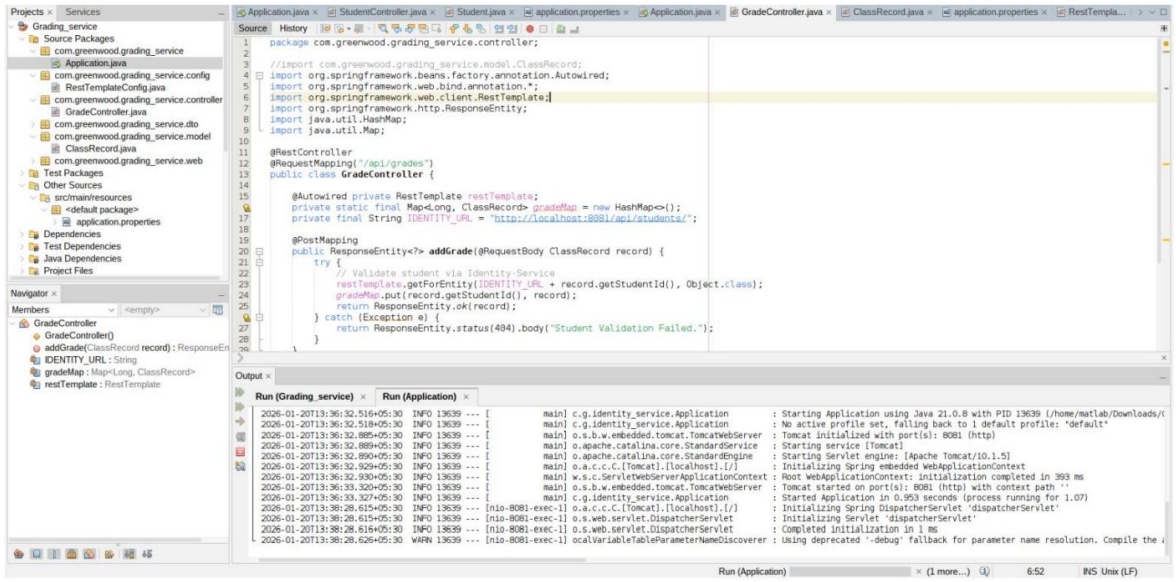


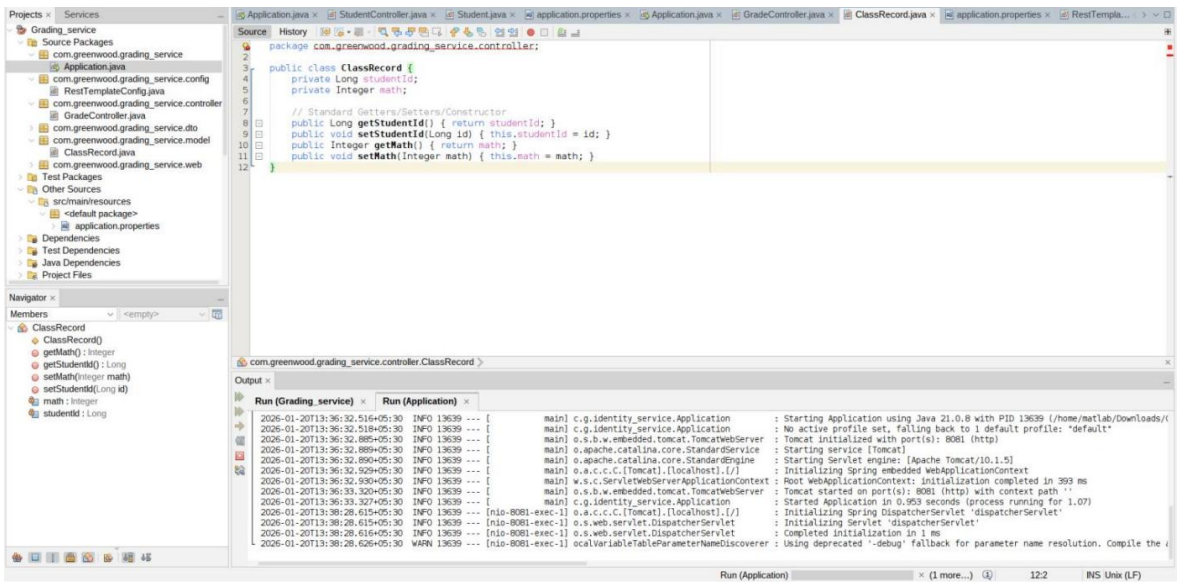
Checking if it runs and has no issues:

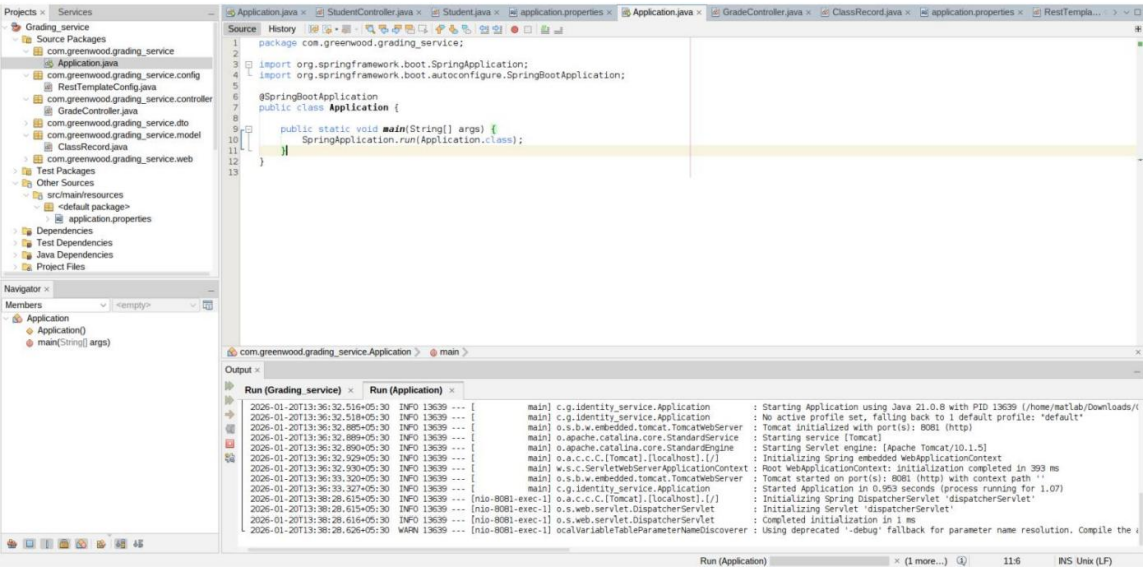




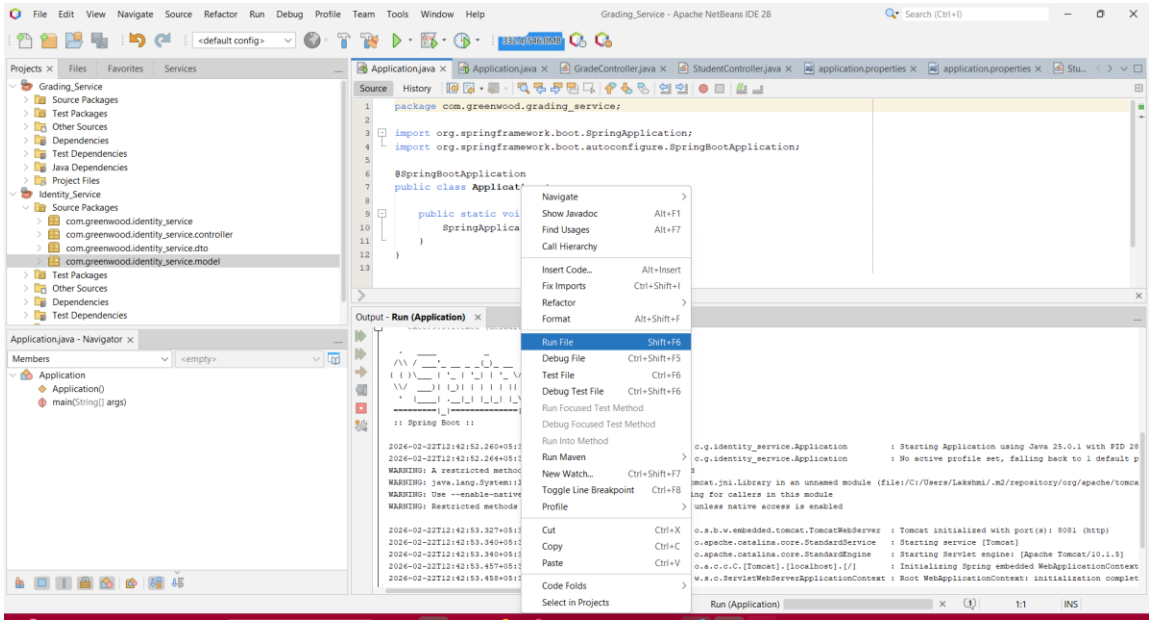
Microservice 2 (Grading service):

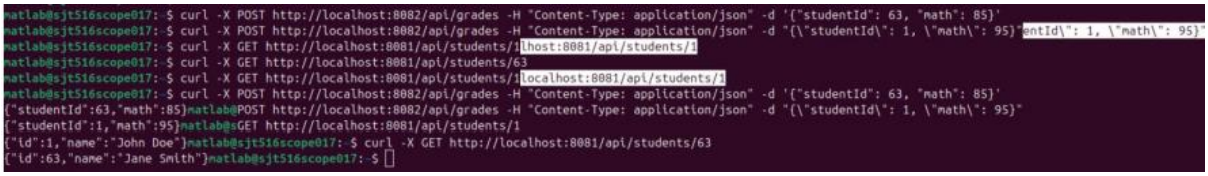




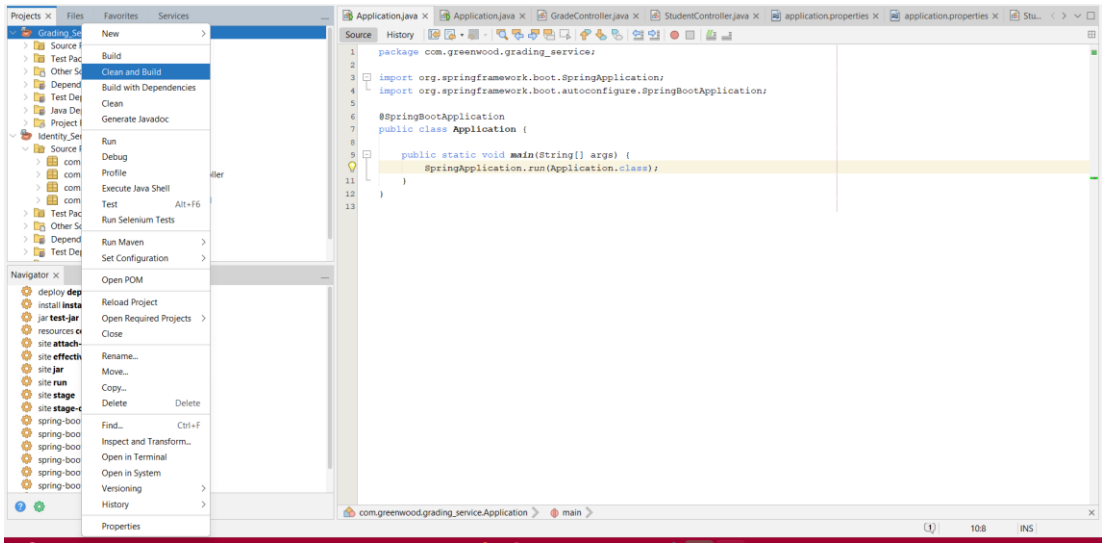


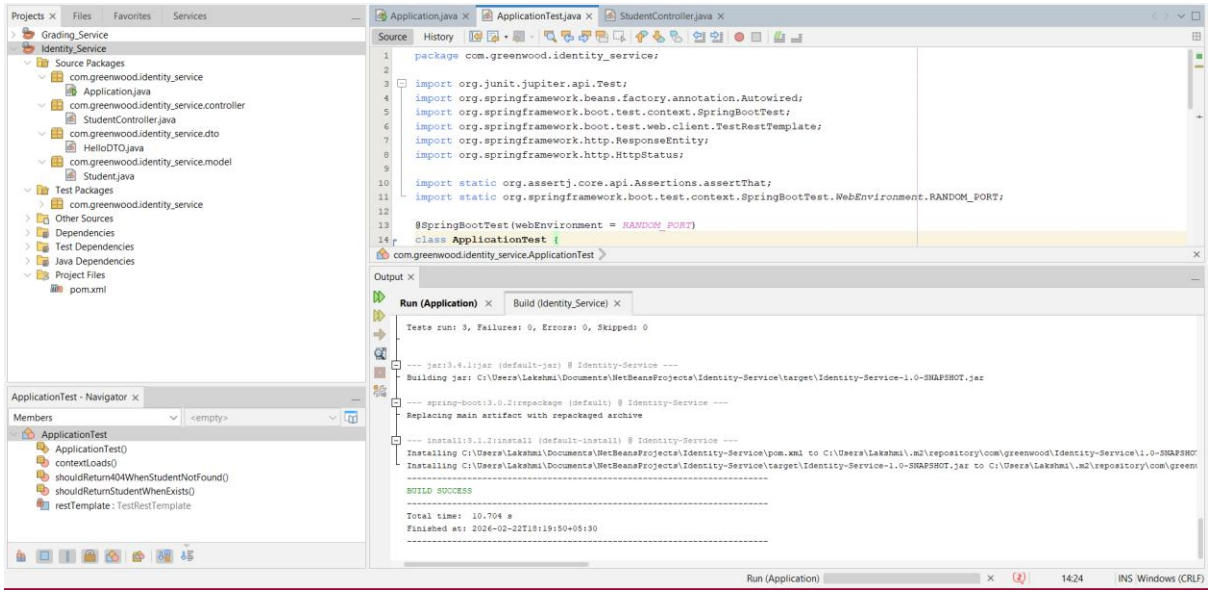
Checking if it runs and has no issues:

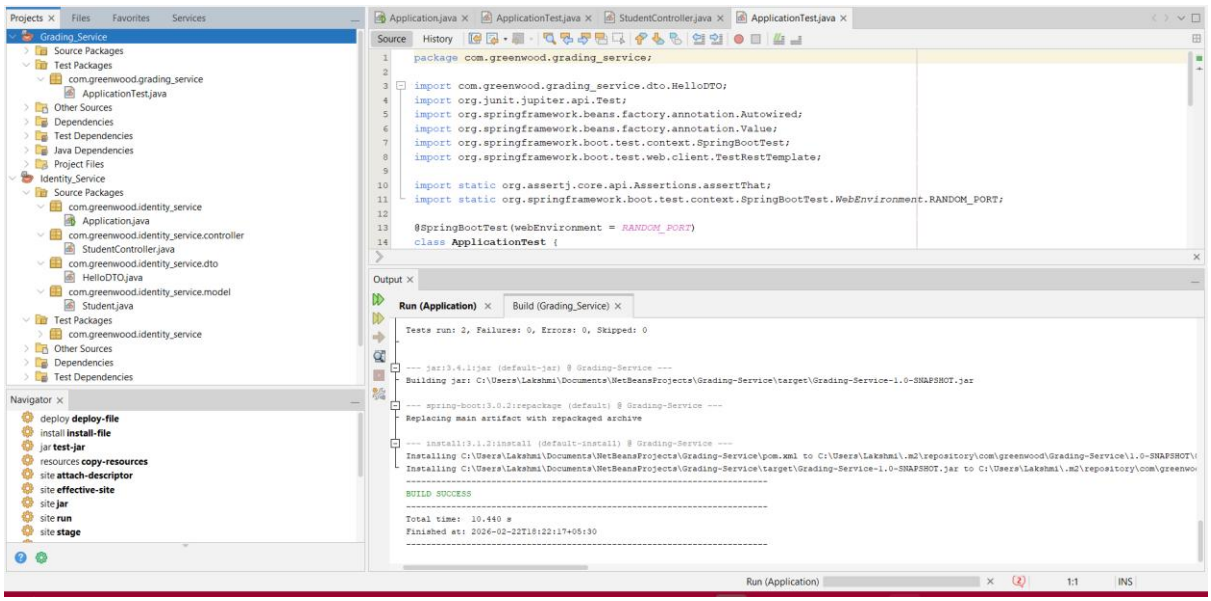




Bundle both services into jar files:







Check if the .jar files are working are working:

```
C:\Users\Lakshmi\Documents\NetBeansProjects\Identity-Service\target>"C:\Program Files\Apache NetBeans\jdk\bin\java.exe"
-jar Identity-Service-1.0-SNAPSHOT.jar

  .   ____          _            __ _ _
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( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.0.2)

2026-02-22T18:32:06.009+05:30  INFO 31236 --- [           main] c.g.identity_service.Application         : Starting Appl
ication using Java 25.0.1 with PID 31236 (C:\Users\Lakshmi\Documents\NetBeansProjects\Identity-Service\target\Identity-S
ervice-1.0-SNAPSHOT.jar started by Lakshmi in C:\Users\Lakshmi\Documents\NetBeansProjects\Identity-Service\target)
2026-02-22T18:32:06.014+05:30  INFO 31236 --- [           main] c.g.identity_service.Application         : No active pro
file set, falling back to 1 default profile: "default"
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.apache.tomcat.jni.Library in an unnamed module (jar:file:/C:/User
s/Lakshmi/Documents/NetBeansProjects/Identity-Service/target/Identity-Service-1.0-SNAPSHOT.jar!/BOOT-INF/lib/tomcat-embe
d-core-10.1.5.jar!/)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-02-22T18:32:07.201+05:30  INFO 31236 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initia
lized with port(s): 8081 (http)
2026-02-22T18:32:07.216+05:30  INFO 31236 --- [           main] o.apache.catalina.core.StandardService   : Starting serv
ice [Tomcat]
2026-02-22T18:32:07.217+05:30  INFO 31236 --- [           main] o.apache.catalina.core.StandardEngine    : Starting Serv
let engine: [Apache Tomcat/10.1.5]
2026-02-22T18:32:07.323+05:30  INFO 31236 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing
```

```
C:\Users\Lakshmi\Documents\NetBeansProjects\Grading-Service\target>"C:\Program Files\Apache NetBeans\jdk\bin\java.exe" -
jar Grading-Service-1.0-SNAPSHOT.jar

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.0.2)

2026-02-22T18:35:01.041+05:30  INFO 24212 --- [           main] c.greenwood.grading_service.Application  : Starting Appl
ication using Java 25.0.1 with PID 24212 (C:\Users\Lakshmi\Documents\NetBeansProjects\Grading-Service\target\Grading-Ser
vice-1.0-SNAPSHOT.jar started by Lakshmi in C:\Users\Lakshmi\Documents\NetBeansProjects\Grading-Service\target)
2026-02-22T18:35:01.046+05:30  INFO 24212 --- [           main] c.greenwood.grading_service.Application  : No active pro
file set, falling back to 1 default profile: "default"
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.apache.tomcat.jni.Library in an unnamed module (jar:file:/C:/User
s/Lakshmi/Documents/NetBeansProjects/Grading-Service/target/Grading-Service-1.0-SNAPSHOT.jar!/BOOT-INF/lib/tomcat-embed-
core-10.1.5.jar!/)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-02-22T18:35:02.410+05:30  INFO 24212 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initia
lized with port(s): 8082 (http)
2026-02-22T18:35:02.426+05:30  INFO 24212 --- [           main] o.apache.catalina.core.StandardService   : Starting serv
ice [Tomcat]
2026-02-22T18:35:02.427+05:30  INFO 24212 --- [           main] o.apache.catalina.core.StandardEngine    : Starting Serv
let engine: [Apache Tomcat/10.1.5]
```

Make the docker file and save in the root folder:

Dockerfile

File Edit View

```
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY ./target/*.jar ./app.jar
EXPOSE 8085
ENTRYPOINT ["java", "-jar", "app.jar"]
```

Build the image for both services:

```
C:\Users\Lakshmi\Documents\NetBeansProjects>docker build -t identity-service:latest ./identity-service
[+] Building 27.2s (9/9) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                  0.2s
 => => transferring dockerfile: 162B                                                                                  0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                            5.1s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io                                               0.0s
 => [internal] load .dockerignore                                                                                     0.1s
 => => transferring context: 2B                                                                                       0.0s
 => [internal] load build context                                                                                     1.2s
 => => transferring context: 24.47MB                                                                                  1.1s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:b624cb9175b71aaeb654dd9def666035332d5abf70318537c  19.4s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:b624cb9175b71aaeb654dd9def666035332d5abf70318537c1  0.1s
 => => sha256:fd88562ce968f9f2742decb2b41610942b1696ea2c5e3d23e5a6d4d60c881435 2.28kB / 2.28kB                       0.6s
 => => sha256:78395afbc5ebce60b7dff0200af60f0f3df93dd724ebb8f148212ed7d9b243e7 159B / 159B                           1.0s
 => => sha256:01d7766a2e4a62b74e0bebf2cd12c47e675e9221174f6570854203e84ffe68b0 29.73MB / 29.73MB                     4.7s
 => => sha256:1bbcf5688c1943d868111317459775672c2eb59c07b04633f4653e36deaac799 145.63MB / 145.63MB                  16.8s
 => => sha256:d10f3ac7b458bcc0853b8cddf5bb91d305f64732b5e46f7897402e9104a2b6c7 22.96MB / 22.96MB                      7.3s
 => => extracting sha256:01d7766a2e4a62b74e0bebf2cd12c47e675e9221174f6570854203e84ffe68b0                            2.0s
 => => extracting sha256:d10f3ac7b458bcc0853b8cddf5bb91d305f64732b5e46f7897402e9104a2b6c7                            1.3s
 => => extracting sha256:1bbcf5688c1943d868111317459775672c2eb59c07b04633f4653e36deaac799                            2.1s
 => => extracting sha256:78395afbc5ebce60b7dff0200af60f0f3df93dd724ebb8f148212ed7d9b243e7                            0.0s
 => => extracting sha256:fd88562ce968f9f2742decb2b41610942b1696ea2c5e3d23e5a6d4d60c881435                            0.0s
 => [2/3] WORKDIR /app                                                                                                0.5s
 => [3/3] COPY ./target/*.jar ./app.jar                                                                               0.2s
 => exporting to image                                                                                                1.5s
 => => exporting layers                                                                                               1.0s
 => => exporting manifest sha256:026a8f5f65a95300f4a443bba4d4f0e85f99d189f1d162d6186eb49e656492d5                    0.0s
 => => exporting config sha256:22d4a1f0143988a1f75604304c0d7baf2ff63b42a15efdb9bfe3337e24cdb355                       0.0s
 => => exporting attestation manifest sha256:d4d4752fc06d38b6eefd6c0b2c804f39ec09635ade343fb38ad02b390c6a9a79         0.0s
 => => exporting manifest list sha256:b0b1d2c9b2f0e9a5f7f7ab9b3875e5f7aaf2c6f3ae708101f1ec2884c45be942               0.0s
 => => naming to docker.io/library/identity-service:latest                                                            0.0s
 => => unpacking to docker.io/library/identity-service:latest                                                         0.2s

C:\Users\Lakshmi\Documents\NetBeansProjects>
```

```
C:\Users\Lakshmi\Documents\NetBeansProjects>docker build -t grading-service:latest ./grading-service
[+] Building 4.0s (8/8) FINISHED                                                                     docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                  0.1s
 => => transferring dockerfile: 162B                                                                                  0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                            1.3s
 => [internal] load .dockerignore                                                                                     0.0s
 => => transferring context: 2B                                                                                       0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:b624cb9175b71aaeb654dd9def666035332d5abf70318537c1a46e61564dbecd  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:b624cb9175b71aaeb654dd9def666035332d5abf70318537c1a46e61564dbecd  0.0s
 => [internal] load build context                                                                                     0.9s
 => => transferring context: 24.47MB                                                                                  0.9s
 => CACHED [2/3] WORKDIR /app                                                                                         0.0s
 => [3/3] COPY ./target/*.jar ./app.jar                                                                               0.1s
 => exporting to image                                                                                                1.4s
 => => exporting layers                                                                                               1.0s
 => => exporting manifest sha256:38bdcedf537bf0514891236b2a68251601903bf14740c649de6d27c69d805720                    0.0s
 => => exporting config sha256:35193a7f05251a59e0cdae8472009a9fafd3be5912bfe609e8b11158bdca17b7                       0.0s
 => => exporting attestation manifest sha256:45492d3a8bb95d71028ff0409e04a7f0540442b0b0e2f6441797e8fead4ad725         0.0s
 => => exporting manifest list sha256:af9bb7770c43bb10fe0ee18d854f3c0a837b0fba57a7cee8b2b5d674c6f98443               0.0s
 => => naming to docker.io/library/grading-service:latest                                                             0.0s
 => => unpacking to docker.io/library/grading-service:latest                                                          0.2s

C:\Users\Lakshmi\Documents\NetBeansProjects>
```

Run the microservices on docker:

```
C:\Users\Lakshmi>docker run -p 8081:8081 identity-service

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 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.0.2)

2026-02-22T17:45:59.406Z  INFO 1 --- [           main] c.g.identity_service.Application
: Starting Application using Java 17.0.18 with PID 1 (/app/app.jar started by root in /app)
2026-02-22T17:45:59.412Z  INFO 1 --- [           main] c.g.identity_service.Application
: No active profile set, falling back to 1 default profile: "default"
2026-02-22T17:46:01.202Z  INFO 1 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer
: Tomcat initialized with port(s): 8081 (http)
2026-02-22T17:46:01.222Z  INFO 1 --- [           main] o.apache.catalina.core.StandardService
: Starting service [Tomcat]
2026-02-22T17:46:01.223Z  INFO 1 --- [           main] o.apache.catalina.core.StandardEngine
: Starting Servlet engine: [Apache Tomcat/10.1.5]
2026-02-22T17:46:01.382Z  INFO 1 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]
: Initializing Spring embedded WebApplicationContext
2026-02-22T17:46:01.387Z  INFO 1 --- [           main] w.s.c.ServletWebServerApplicationContext
: Root WebApplicationContext: initialization completed in 1865 ms
```

```
PS C:\Users\Lakshmi> docker run -p 8082:8082 grading-service

  .   ____          _            __ _ _
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( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
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 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.0.2)

2026-02-22T17:47:17.772Z  INFO 1 --- [           main] c.greenwood.grading_service.Application
: Starting Application using Java 17.0.18 with PID 1 (/app/app.jar started by root in /app)
2026-02-22T17:47:17.778Z  INFO 1 --- [           main] c.greenwood.grading_service.Application
: No active profile set, falling back to 1 default profile: "default"
2026-02-22T17:47:19.667Z  INFO 1 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer
: Tomcat initialized with port(s): 8082 (http)
2026-02-22T17:47:19.686Z  INFO 1 --- [           main] o.apache.catalina.core.StandardService
: Starting service [Tomcat]
2026-02-22T17:47:19.686Z  INFO 1 --- [           main] o.apache.catalina.core.StandardEngine
: Starting Servlet engine: [Apache Tomcat/10.1.5]
2026-02-22T17:47:19.834Z  INFO 1 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]
: Initializing Spring embedded WebApplicationContext
2026-02-22T17:47:19.836Z  INFO 1 --- [           main] w.s.c.ServletWebServerApplicationContext
: Root WebApplicationContext: initialization completed in 1950 ms
```

Intercommunication between the microservices after running it on Docker:

```
C:\Users\Lakshmi>curl -X GET http://localhost:8081/api/students/1
{"id":1,"name":"John Doe"}
C:\Users\Lakshmi>curl -X GET http://localhost:8081/api/students/63
{"id":63,"name":"Jane Smith"}
```

```
C:\Users\Lakshmi>curl -X POST http://localhost:8082/api/grades -H "Content-Type: application/json" -d "{\"studentId\": 1
, \"math\": 95}"
Student Validation Failed.
C:\Users\Lakshmi>
```